Lab 4 — DP Visualizations

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Specialisation: BCA — Artificial Intelligence and Data Science **Section:** A

Lab: Lab 4 — Smart Optimization Assistant using Dynamic Programming

DP tables, optimization outputs, runtime and memory experiments

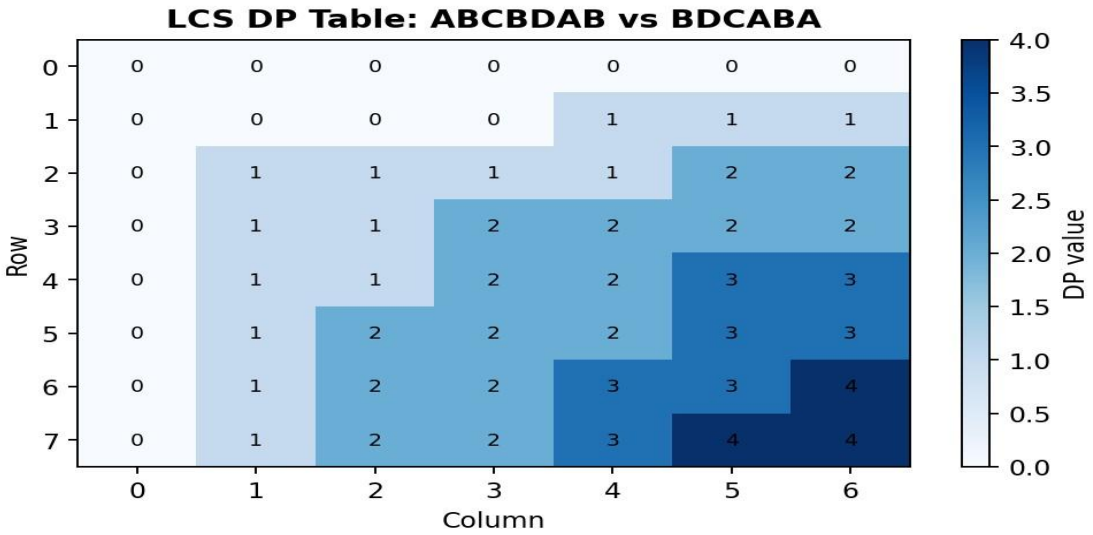
Sample optimization results

LCS: ABCBDAB and BDCABA → length 4, one valid LCS: BCBA.

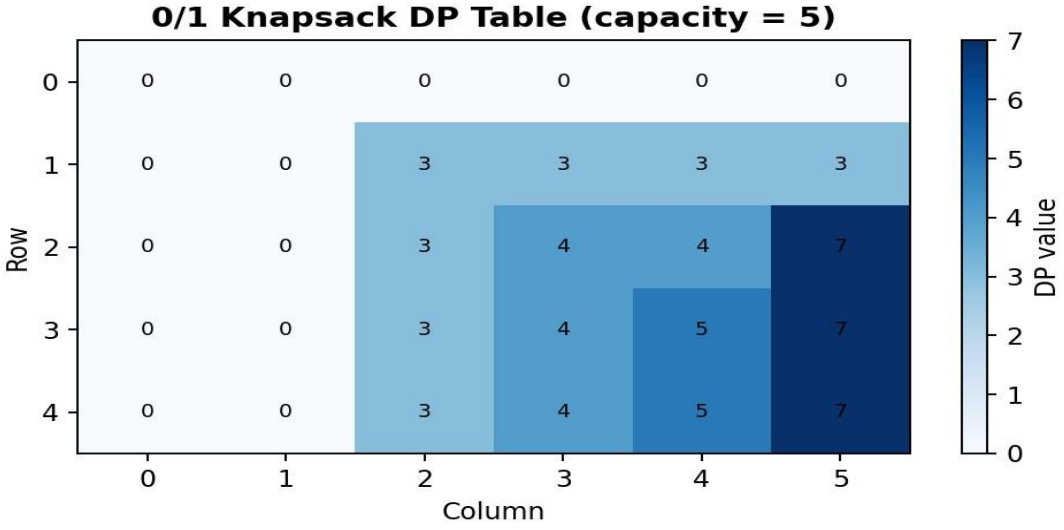
0/1 Knapsack: weights [2, 3, 4, 5], values [3, 4, 5, 6], capacity 5 → maximum profit 7, selected 1-based item numbers [1, 2].

Matrix Chain: dimensions [10, 30, 5, 60] → minimum 4500 scalar multiplications; optimal order ((A1 × A2) × A3).

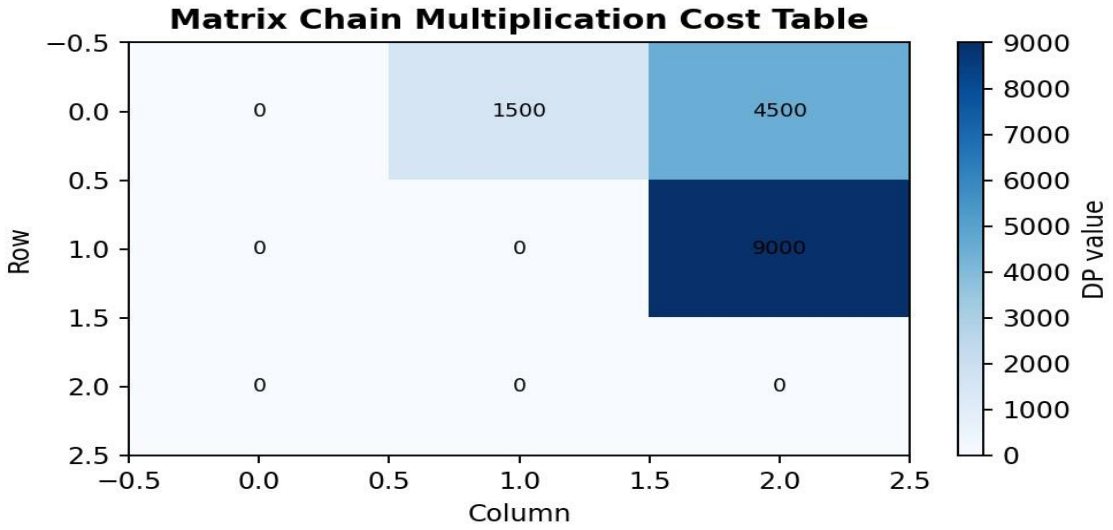
LCS DP table



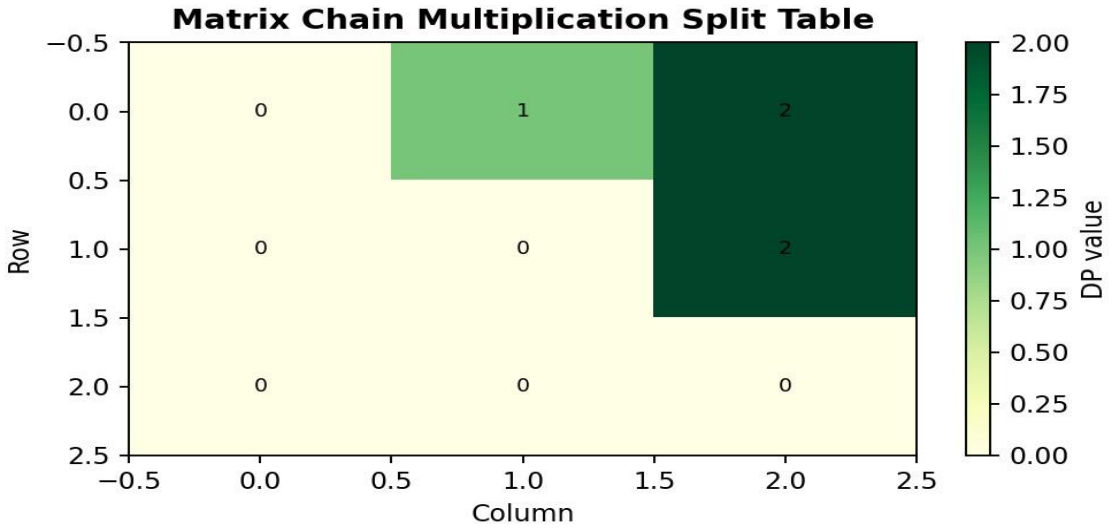
0/1 Knapsack DP table



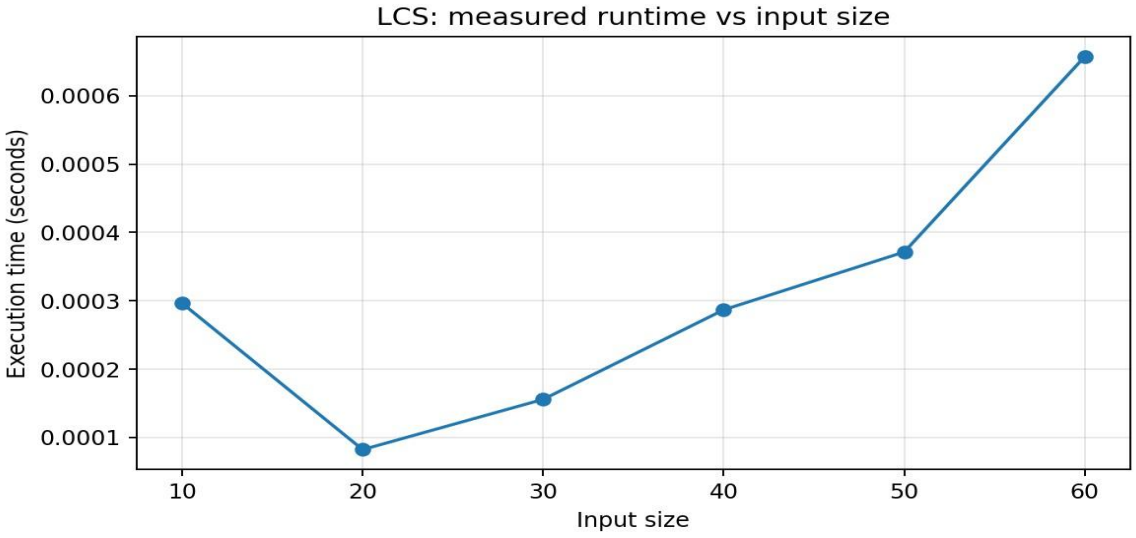
Matrix Chain cost table



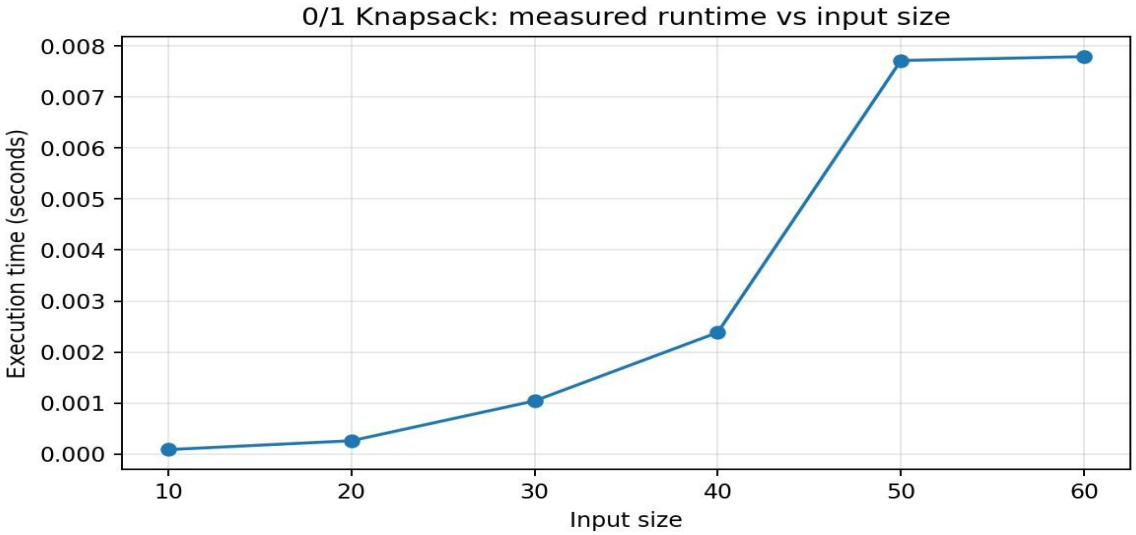
Matrix Chain split table



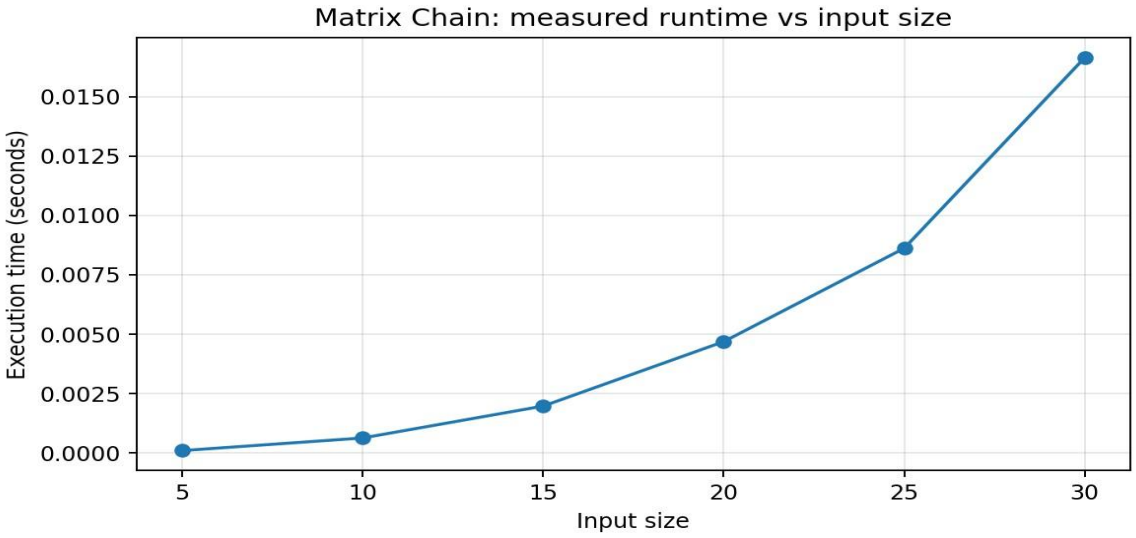
LCS runtime growth



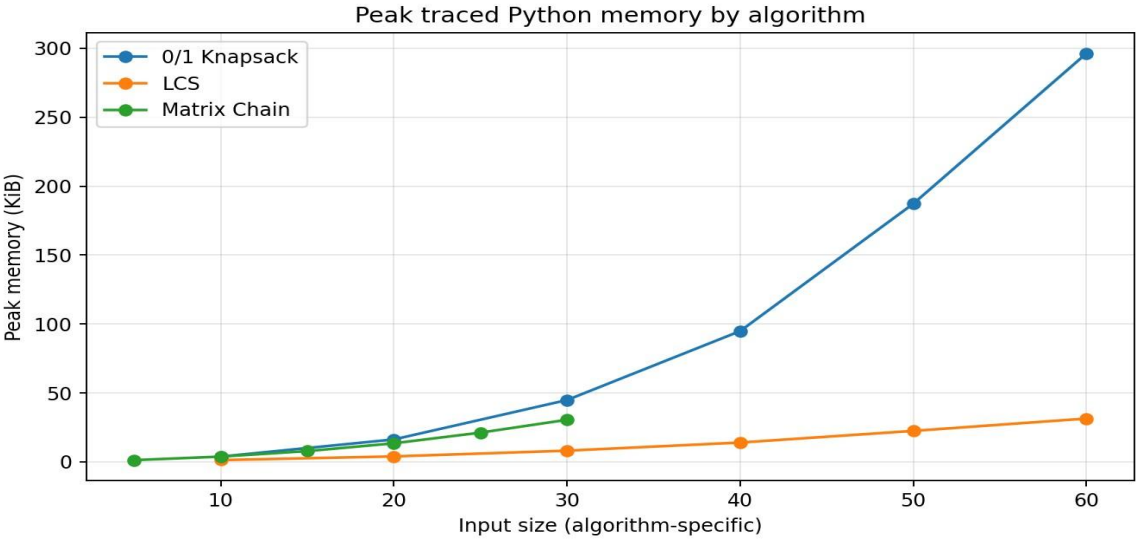
Knapsack runtime growth



Matrix Chain runtime growth



Memory comparison



Experimental observations Experimental observations

• LCS computes a rectangular table. Its state count grows with the product of the two sequence lengths.

• 0/1 Knapsack runtime depends on both item count and capacity; its O(nW) bound is pseudo-polynomial in numeric capacity W.

• Matrix Chain Multiplication stores O(n^2) interval states and examines possible split positions, giving O(n^3) time.

• Traced memory is a Python-level measurement and is not identical to total process RAM.

• Measured values are hardware- and interpreter-dependent. The CSV is an experimental sample, not a universal benchmark.